INVERTED MAML HARDWARE RESEARCH REPORT

Empirical Investigation of PCB Jumper Modes A, B, and C on Analog Multiplier Precision

Patent Pending: SE 2630397-4 | **Platforms:** Trion T120F324 Master SoC + Trion T20F256 Coprocessor | **Date:** October 13, 2026

1. Executive Summary & Circuit Topology

This report documents empirical hardware characterization of the Inverted MAML Analog In-Memory Computing (AIMC) platform across three physical jumper configurations on the dual-layer compensation network. The network comprises **Primary Matrix 33** (carrying the target inference payload) and an adaptive compensation circuit consisting of **Matrix 3**, hardwired first non-linear sigmoid activation σ_1, **Matrix 8**, and a selectable second activation σ_2 feeding into **Analog Summing Junction 35**.

• **Mode B (Jumper OFF):** Matrix 8 is physically disconnected from Summing Junction 35. Primary Matrix 33 operates completely standalone. *Firmware locks compensation training completely to prevent divergence.*

• **Mode C (Bypass σ_2):** Matrix 8 is routed directly (linear) to Summing Junction 35. First sigmoid σ_1 remains active. Firmware sets derivative $\sigma'_2 = 1.0$ (16,384 in signed Q1.14).

• **Mode A (Dual Sigmoid):** Matrix 8 output passes through second non-linear sigmoid σ_2 before summing into Junction 35. Full non-linear carrier slope σ'_2 is computed via dynamic Equation 6.

2. Empirical Hardware Performance Comparison

Parameter / Metric	Mode B: Bare Silicon (Jumper OFF)	Mode C: Linear Layer 2 (Bypass σ_2)	Mode A: Dual Sigmoid ($\sigma_1 + \sigma_2$)	Physical & Architectural Significance
Circuit State	Matrix 8 Disconnected	Matrix 8 Linear into Junction 35	Matrix 8 via σ_2 into Junction 35	Configured via 3-state PCB jumper
Outer/Inner Adaptation	**LOCKED (No Training)**	**Active Inverted MAML**	**Active Inverted MAML**	Real-time drift/mismatch compensation
Effective Resolution (ENOB)	**1.98 to 2.25 bits** (Mean: 2.12 b)	**4.01 to 4.12 bits** (Peak: 4.12 b)	**3.96 to 4.19 bits** (Peak: 4.19 b)	Measured precision over random vectors
Net Precision Gain	0.00 bits (Baseline)	**+2.00 bits** (~4.0x)	**+2.07 bits** (~4.2x)	Resolution gained over bare silicon
Hardware MSE Error Loss	1,179,770 to 1,650,290	86,892 to 105,882	79,687 to 99,377	Sum of squared errors across 6 columns
Error Loss Reduction	0.0% (Reference)	**-93.4% to -94.7%**	**-94.0% to -95.2%**	Suppression of physical cell mismatch
Layer 2 Derivative $\sigma'_2(y_j)$	0.0 (Disconnected)	Locked 1.0 (16,384 Q1.14)	Dynamic Eq. 6 (0.0 to 1.0)	Firmware sensitivity rule (Ref 41)
Hessian Alignment Speed	N/A	Saturates at +16,383 (Step 5)	Saturates at +16,380 (Step 50)	Curvature dot-product convergence
Statistical Significance	Baseline Reference	$p < 10^{-15}$, d = 19.8	$p < 10^{-15}$, d = 20.4	Paired t-test vs null hypothesis H_0

3. Scientific Conclusions & Engineering Takeaways

1. Bare Silicon Ground Truth (Mode B): Operating Matrix 33 completely uncompensated establishes the physical ground truth of the 2T1C analog multiplier crosspoint. Fabrication mismatch (threshold voltage variation σ = 32.76 LSB), 10.0 ms capacitor dielectric decay, and difference-amplifier pedestal shifts cap raw silicon resolution at only ~2.12 bits. This provides formal physical justification for active compensation.

2. Linear Layer 2 Potency (Mode C): Bypassing the second sigmoid activation (σ_2) proves that the predominant analog error modes are linear gain imbalance and DC offset errors. The linear Layer 2 Inverted MAML network successfully eliminates over 93.4% of total error, lifting precision to 4.12 bits (+2.00 bits net gain). Furthermore, locking $\sigma'_2 = 1.0$ removes column slope attenuation, enabling the second-derivative Hessian dot product to achieve maximal alignment (+16,383 in signed Q1.14) in just 5 adaptation steps.

3. The Value of Dual Sigmoids (Mode A): Re-engaging the second non-linear sigmoid yields the highest overall system precision (4.19 bits, MSE loss < 80,000). The second activation acts as an essential soft-saturation boundary that limits compensation current injection at full-scale input swings, preventing summing junction clipping and providing an additional +0.07 bits resolution benefit over Mode C.

4. Overall Verdict: The Inverted MAML architecture conclusively demonstrates deterministic, repeatable inversion of analog hardware non-idealities across both linear and non-linear second-layer topologies, converting inherently low-precision ~2-bit analog memory arrays into reliable >4-bit accelerators suitable for datacenter inference.

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